

Paper: 1MA1/2H																				
Question	Answer	Mark	Mark scheme	Additional guidance																
19	60 and reason	B1	for 432.5 or 437.5 or 9.75 or 9.85 or 2.45 or 2.55																	
		M1	for a correct process to find a bound for T eg [LB of w] $\div$ [UB of $a - \text{LB of } c$] where 432.5 ,, [LB of w] $<$ 435 and $9.8 < [\text{UB of } a]$,, 9.85 and 2.45 ,, [LB of c] $<$ 2.5 or [UB of w] $\div$ [LB of $a - \text{UB of } c$] where $435 < [\text{UB of } w]$,, 437.5 and 9.75 ,, [LB of a] $<$ 9.8 and $2.5 < [\text{UB of } c]$,, 2.55	<table><tr><th>Lette r</th><th>Given</th><th>LB</th><th>UB</th></tr><tr><td>w</td><td>435</td><td>432.5</td><td>437.5</td></tr><tr><td>a</td><td>9.8</td><td>9.75</td><td>9.85</td></tr><tr><td>c</td><td>2.5</td><td>2.45</td><td>2.55</td></tr></table>	Lette r	Given	LB	UB	w	435	432.5	437.5	a	9.8	9.75	9.85	c	2.5	2.45	2.55
		Lette r	Given	LB	UB															
		w	435	432.5	437.5															
		a	9.8	9.75	9.85															
		c	2.5	2.45	2.55															
M1	for a correct process to find both LB and UB bound for T eg [LB of w] $\div$ [UB of $a - \text{LB of } c$] where 432.5 ,, [LB of w] $<$ 435 and $9.8 < [\text{UB of } a]$,, 9.85 and 2.45 ,, [LB of c] $<$ 2.5 and [UB of w] $\div$ [LB of $a - \text{UB of } c$] where $435 < [\text{UB of } w]$,, 437.5 and 9.75 ,, [LB of a] $<$ 9.8 and $2.5 < [\text{UB of } c]$,, 2.55	UB <table><tr><th>Lette r</th><th>Given</th><th>LB</th><th>UB</th></tr><tr><td>w</td><td>435</td><td>432.5</td><td>437.5</td></tr><tr><td>a</td><td>9.8</td><td>9.75</td><td>9.85</td></tr><tr><td>c</td><td>2.5</td><td>2.45</td><td>2.55</td></tr></table>	Lette r	Given	LB	UB	w	435	432.5	437.5	a	9.8	9.75	9.85	c	2.5	2.45	2.55		
Lette r	Given	LB	UB																	
w	435	432.5	437.5																	
a	9.8	9.75	9.85																	
c	2.5	2.45	2.55																	
A1	(dep on all previous marks) for 58.44(5...) and 60.76(3...) with both values clearly coming from working with correct values	LB <table><tr><th>Lette r</th><th>Given</th><th>LB</th><th>UB</th></tr><tr><td>w</td><td>435</td><td>432.5</td><td>437.5</td></tr><tr><td>a</td><td>9.8</td><td>9.75</td><td>9.85</td></tr><tr><td>c</td><td>2.5</td><td>2.45</td><td>2.55</td></tr></table>	Lette r	Given	LB	UB	w	435	432.5	437.5	a	9.8	9.75	9.85	c	2.5	2.45	2.55		
Lette r	Given	LB	UB																	
w	435	432.5	437.5																	
a	9.8	9.75	9.85																	
c	2.5	2.45	2.55																	
C1	for 60 from 58.44... and 60.76... and statement that both LB and UB round to 60																			
				Accept bounds rounded or truncated to at least 4 sf																